Dynamics-Inspired Dual-Representation Learning for Distributed Acoustic Sensing Event Classification

***Abstract*—Distributed acoustic sensing (DAS) supports vibration monitoring along optical fibers, but noise can obscure features needed for event recognition. This paper presents PG-ZGCA, a physics-guided network that combines a measured waveform with a dynamics-inspired auxiliary representation through zero-initialized gated cross-attention. An auxiliary convolutional network is pretrained to estimate a mass–spring–damper-inspired target from synthetically corrupted recordings. Two independent ResNet18 encoders extract features from the power spectrograms of the input and auxiliary output. Eight-head cross-attention uses auxiliary features as queries and measured-signal features as keys and values; a zero-initialized scalar controls the attention residual before layer normalization and classification. On a six-class DAS collection, the reported single-run validation results at an added-noise signal-to-noise ratio of -10 dB were 92.97% accuracy, 73.19% macro recall, and 74.09% macro F1. Evaluation used an 80/20 random sample split, with validation reuse and full-collection auxiliary pretraining. The findings characterize performance under synthetic noise within this collection. The auxiliary representation serves as an equivalent signal prior, with its physical interpretation conditional on the assumed measurement model.**

***Index Terms*—Distributed acoustic sensing, event recognition, physics-guided learning, cross-attention, zero-initialized gating, noise robustness.**

I. Introduction

Distributed acoustic sensing (DAS) uses optical fibers as distributed sensing elements for vibration monitoring. It supports applications in geophysical observation and infrastructure monitoring [1], [2]. Event recognition translates these measurements into labels that can support subsequent interpretation. A practical difficulty is that recorded windows contain background fluctuations as well as event-related components. A classifier must retain useful temporal and spectral structure as additional noise changes the input distribution.

Deep neural networks provide several ways to model DAS recordings. Residual convolutional networks learn hierarchical representations [3], and recurrent architectures have been investigated for DAS event detection and monitoring [4]. Self-supervised denoising has also been studied for offshore DAS measurements [5]. These approaches motivate learning representations that remain useful under noise. For hybrid models, interpreting recognition gains requires distinguishing the roles of the physical prior, model capacity, and training protocol.

Physics-informed learning provides a way to introduce governing relationships into neural estimation. For example, wave-propagation and inversion studies have used physics-informed neural networks (PINNs) [6]. In event classification, a physical model can instead define an auxiliary representation that conditions the classifier. The usefulness of that representation depends on both its physical assumptions and the mechanism used to combine it with measured features. Cross-attention offers a learned interface between feature sequences, while the broader multimodal literature documents challenges associated with noisy and unequal-quality inputs [7]. Here, both branches originate from the same DAS waveform and should be understood as two representations of one measurement.

This paper studies a physics-guided zero-initialized gated cross-attention network, termed PG-ZGCA. An auxiliary one-dimensional convolutional network learns a dynamics-inspired transformation of the waveform. Its output and the original waveform are converted to power spectrograms and encoded by independent ResNet18 branches. Auxiliary features query raw features through multi-head cross-attention, and a learned scalar controls the resulting residual contribution. Zero-initialized residual gates have precedents in ReZero [8], and gated cross-attention has been used in Flamingo [9]. The present design applies these principles to the combination of a dynamics-inspired auxiliary representation and DAS features.

The work makes three specific contributions. First, it formulates an auxiliary pretraining target using detrending and discrete integrations, and identifies its role as an equivalent signal prior. Second, it specifies a dual-encoder fusion architecture in which auxiliary queries condition raw-feature attention and the attention residual starts at zero. Third, it reports noise-dependent validation results and comparisons of architectural variants on a six-class DAS collection, with explicitly defined metrics and evaluation conditions. The experimental evidence concerns the available collection and synthetic additive noise; independent field generalization remains a separate validation objective.

Section II discusses related work. Section III specifies the model and pretraining objective. Section IV describes the data, implementation, and reported results. Section V concludes the paper.

II. Related Work

A. DAS Recognition and Learned Representations

DAS measurements have been used for seismic observation and distributed monitoring [1], [2]. Learning methods operate on waveforms, spectral representations, or spatiotemporal

inputs. RNN-DAS investigated recurrent learning for volcano-tectonic event detection and monitoring [4]; masked autoencoders have been studied for offshore DAS denoising and earthquake detection [5]. These tasks differ from the six-class window recognition considered here, so their results provide methodological context rather than directly comparable benchmark scores.

Residual networks are a standard feature-extraction architecture [3]. For time-series classification, InceptionTime explores multi-scale convolutional representations [10], while attention-based multiple-instance learning aggregates a sequence of learned instances [11]. The comparison implementations in this study include simplified convolutional and recurrent models based on these ideas. Their local architecture definitions, rather than their abbreviated names alone, determine the scope of the comparison.

B. Physics-Guided Auxiliary Learning

PINNs incorporate physical relationships into neural optimization, including in wave modeling and full-waveform inversion [6]. In the present implementation, the auxiliary network predicts a target derived from a second-order dynamics-inspired operator. It is a feed-forward convolutional estimator trained by target matching and amplitude regularization. This differs from directly solving a spatial wave equation or enforcing a calibrated DAS observation model.

A key distinction is between a physical measurement and an equivalent representation. Interpreting a DAS recording as physical acceleration in SI units requires a calibrated measurement model. Applying a mass–spring–damper expression to a normalized sensor trace therefore defines an equivalent prior unless the observation transfer function and coefficients have been independently calibrated. The formulation below retains that distinction and provides the exact discrete operations used to construct the auxiliary target.

C. Attention and Residual Gating

Cross-attention learns correspondences between query features and key/value features. Related feature-fusion work has applied cross-attention to bearing diagnosis [12]. A learned attention score is a statistical feature similarity; it is not by itself a test of whether a component satisfies a physical law. Moreover, fusion performance can depend on the quality of each input representation [7].

ReZero introduced learnable residual weights initialized to zero [8]. Flamingo used gated cross-attention when conditioning an existing model on additional representations [9]. PG-ZGCA adopts a zero-initialized scalar for the attention residual. Its post-addition layer normalization remains active at initialization, which determines the precise reduction of the model when the gate is zero.

III. Proposed Methodology

Fig. 1 shows the architecture. Let $x \in \mathbb{R}^L$ denote a single-channel window after division by 5000, with $L = 5000$. The model produces class logits from the measured window and an auxiliary output $g_\theta(x)$. The formulation below describes the zero-gated ResNet18 implementation used in the comparison and architectural-variant scripts.

A. Dynamics-Inspired Auxiliary Target

A mass–spring–damper expression motivates an equivalent feature transformation:

$$F_{\mathrm{eq}}(t) = ma(t) + cv(t) + kd(t). \tag{1}$$

Equation (1) describes second-order dynamics when a, v, and d denote physical acceleration, velocity, and displacement, respectively. In this implementation, the normalized DAS trace is treated as an acceleration-like input. The output is thus an equivalent force-like signal, rather than a calibrated external force. The operator uses an integration interval of $\Delta t = 0.001$, corresponding to an assumed sampling rate of 1 kHz. The actual acquisition sampling rate remains unverified.

The target construction removes linear trends before and after integration. For any length-L sequence y, let u_i be equally spaced on $[-1, 1]$. The detrending operator is

$$\begin{aligned} \mathcal{D}(y)_i &= y_i - \bar{y} - \beta_y u_i, \\ \beta_y &= \frac{\sum_{i=0}^{L-1} u_i (y_i - \bar{y})}{\sum_{i=0}^{L-1} u_i^2}. \end{aligned} \tag{2}$$

Using the detrending operator in (2) and writing $\mathcal{I}(y)_i = \Delta t \sum_{j=0}^{i} y_j$, the discrete target is

$$\begin{aligned} &a = \mathcal{D}(x), \qquad v = \mathcal{D}(\mathcal{I}(a)), \\ &d = \mathcal{D}(\mathcal{I}(v)), \quad F_{\mathrm{eq}} = ma + |c_r| v + |k_r| d. \end{aligned} \tag{3}$$

Here, $m = 1$ is fixed, while c_r and k_r are learnable scalars initialized to 1 and 10. Their absolute values provide non-negative coefficients. Both coefficients are optimized jointly with the auxiliary CNN, so the target in (3) also changes during pretraining. This is a learned equivalent target and not independent force ground truth.

B. Auxiliary Network and Pretraining

The auxiliary CNN has four one-dimensional convolutions with channel widths $1 \to 16 \to 32 \to 16 \to 1$, kernel size 15, and padding 7. GELU activations follow the first three convolutions. The forward mapping is

$$g_\theta(x) = W_4 * \sigma\big(W_3 * \sigma(W_2 * \sigma(W_1 * x))\big), \tag{4}$$

In (4), the convolutional layers include biases. The architecture is a sequential CNN with no internal residual connections.

During pretraining, Gaussian noise is added to each normalized recording:

$$\widetilde{x} = x + n, \quad n_i \sim \mathcal{N}\left(0, \frac{L^{-1} \sum_j x_j^2}{10^{s/10}}\right), \quad s \sim \mathcal{U}(-10, 40). \tag{5}$$

The noisy sequence is used by g_θ, while the corresponding recording before added noise is used in (3). The latter retains

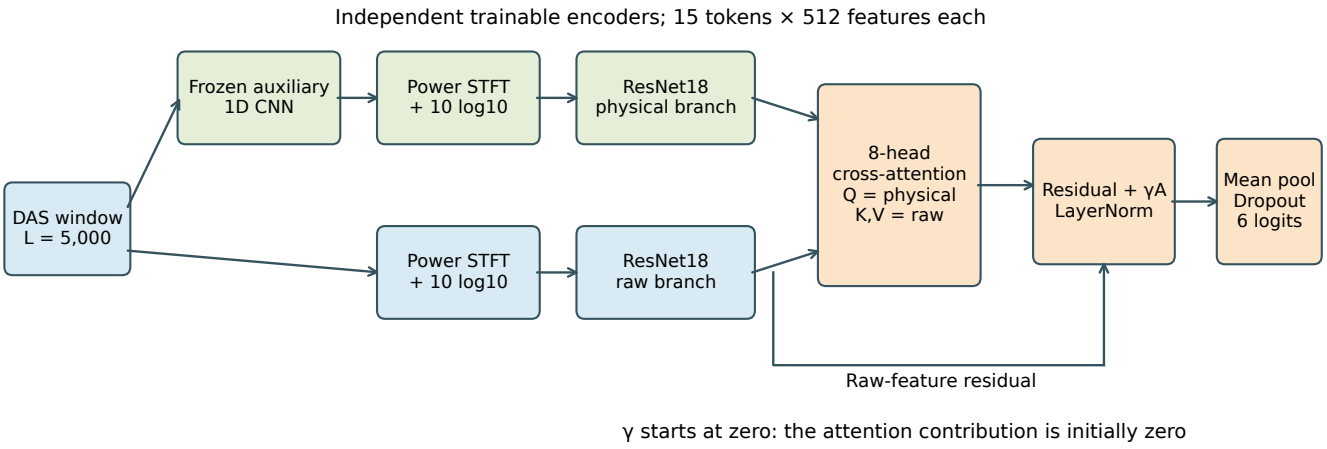


Fig. 1. PG-ZGCA combines two representations of the same DAS window. The auxiliary CNN is frozen during the classifier training specified by the current implementation; both ResNet18 encoders and the fusion layers are trainable. The raw-feature residual bypasses cross-attention, and a mean-pooling classification head produces six logits.

its original measurement noise; it is not a separately measured clean signal. The objective is

$$\begin{aligned} \mathcal{L}_{\text{aux}} &= \mathcal{L}_{\text{target}} + 2\mathcal{L}_1, \\ \mathcal{L}_{\text{target}} &= \frac{1}{BL}\sum_{b,i}\left(g_\theta(\widetilde{x}^{(b)})_i - F^{(b)}_{\text{eq},i}\right)^2, \\ \mathcal{L}_1 &= \frac{1}{BL}\sum_{b,i}|g_\theta(\widetilde{x}^{(b)})_i|. \end{aligned} \tag{6}$$

The first term in (6) matches the learned dynamics-inspired target. The second is an output-amplitude penalty that encourages sparsity across samples rather than explicitly penalizing high-frequency components. The pretraining script uses Adam with learning rate 10^{-3}, batch size 64, and five epochs. During classifier training, the auxiliary CNN is loaded from a pretrained checkpoint and evaluated without gradients. Its scalar coefficients are used in pretraining target construction, while inference uses the CNN output alone.

C. Power Spectrograms and Independent Encoders

Both the classifier input and its auxiliary output are mapped to power spectrograms. Here, x denotes the waveform presented to the classifier, including added noise when applicable; the pretraining notation in (5) distinguishes the stored waveform from its noisy counterpart. For the centered, reflection-padded waveform $\bar{x}$, the transform is

$$\begin{aligned} Z_x(t,k) &= \sum_{n=0}^{255}\bar{x}[64t+n]w[n]e^{-j2\pi kn/256}, \\ S_x(t,k) &= 10\log_{10}\left(\max\{|Z_x(t,k)|^2, 10^{-10}\}\right), \end{aligned} \tag{7}$$

where w is a Hann window. The implementation uses a 256-point FFT, a hop of 64, power exponent 2, and the power-mode decibel transform with reference 1 and no top-dB clipping. This produces 129×79 bins for a 5000-sample input. The frequency axis is linear; taking magnitude squared and applying a logarithm are nonlinear operations.

Two independently parameterized ResNet18 encoders, initialized without pretrained image weights, process the two spectrograms [3]. Each encoder replaces the RGB input convolution with a single-channel 7×7 convolution of stride 2 and padding 3. The original global pooling and classification layers are removed. Each output has shape $512 \times 5 \times 3$ and is flattened into

$$R = \text{Flatten}(H_r(S_x))^T, \quad P = \text{Flatten}(H_p(S_{g_\theta(x)}))^T, \tag{8}$$

In (8), $R, P \in \mathbb{R}^{15\times 512}$. The encoders are trainable even though the auxiliary CNN is frozen. The implementation adds no explicit positional embeddings to these tokens.

D. Cross-Attention and Zero-Initialized Residual

The attention layer uses eight heads, with feature dimension 64 per head. For head h, the learned projections are

$$Q_h = PW_h^Q + b_h^Q, \quad K_h = RW_h^K + b_h^K, \quad V_h = RW_h^V + b_h^V. \tag{9}$$

The head output and merged output are

$$\begin{aligned} A_h &= \text{Softmax}\left(\frac{Q_h K_h^T}{\sqrt{64}}\right)V_h, \\ A &= \text{Concat}(A_1, \ldots, A_8)W^O + b^O. \end{aligned} \tag{10}$$

Equations (9) and (10) use auxiliary features as queries, while the measured-signal branch supplies keys and values. The projected similarities provide classification-driven weights over raw features, without an explicit physical-consistency constraint.

A learnable scalar γ controls the residual contribution:

$$X = \text{LayerNorm}(R + \gamma A), \qquad \gamma\big|_{\text{init}} = 0. \tag{11}$$

TABLE I
LABEL MAPPING AND SINGLE-CHANNEL SAMPLE COUNTS.

ID	Class	Windows	Samples
0	Mouse biting	974	2922
1	No event	14,370	189,470
2	Manual knocking	7878	39,390
3	Door operation	2938	14,690
4	Manhole-cover operation	7441	37,205
5	Excavation	56,647	283,235
	Total	90,248	566,912

At initialization, (11) reduces exactly to $X = \mathrm{LayerNorm}(R)$. The attention term is initially zero, but layer normalization remains active. The initial forward mapping therefore uses a normalized raw-feature path. This initialization property applies to the forward mapping; it provides no performance lower bound or guarantee of subsequent optimization stability. The gate is an unconstrained scalar and can learn either sign.

The classification head averages tokens and applies dropout with probability 0.5 followed by a linear map:

$$\ell = W_c \,\mathrm{Dropout}\left(\frac{1}{15}\sum_{i=1}^{15} X_i\right) + b_c, \quad \widehat{y} = \arg\max_j \ell_j. \tag{12}$$

The classifier is trained using the six logits in (12) and cross-entropy loss with label smoothing 0.1. The archived full checkpoint contains approximately 23.41 million parameters, of which 23.40 million are trainable under the current freezing configuration.

IV. EXPERIMENTS AND RESULTS

A. Dataset Construction and Labels

The DAS collection consists of waveform files organized by event type. The preprocessing script partitions each multichannel recording into nonoverlapping 5000-sample windows. It assigns the directory label unless the mean absolute amplitude of a window is less than 0.9 times the source recording's global mean absolute amplitude and its maximum absolute amplitude is below 5000. When both conditions hold, the window is assigned to the no-event class. The two conditions are evaluated across each multichannel window, and the resulting window label is assigned to every channel when the window is unpacked. These are heuristic window labels rather than separately verified event onset annotations.

The final dataset contains 90,248 multichannel windows from 17 distinct source filenames. Stored tensor shapes comprise 80,566 windows with five channels, 2180 with three channels, and 7502 with 21 channels. The classifier loader treats each channel as a separate sample and divides its values by 5000, producing 566,912 single-channel examples. Table I gives the exact class distribution obtained from the stored labels and tensor shapes. Counts refer to channel-windows and should not be interpreted as numbers of independent physical events.

B. Training and Evaluation Protocol

The classification scripts set `torch.manual_seed(42)` and randomly split the unpacked single-channel examples into 80% training and 20% validation subsets. For the supplied dataset, these contain 453,529 and 113,383 samples, respectively. This is a sample-level split with no grouping by source recording, acquisition date, or original multichannel window. Channels from a common window and windows from the same recording can therefore appear in both subsets.

The full-model and comparison training scripts specify AdamW with a learning rate of 5×10^{-4}, weight decay 10^{-4}, batch size 64, and 30 epochs. Each training example receives AWGN with an SNR sampled uniformly between -10 and 40 dB according to (5). Cross-entropy uses label smoothing 0.1, and the checkpoint with the highest validation accuracy on inputs without additional AWGN is saved. Training uses a fixed learning rate for all 30 epochs, without cosine scheduling or patience-based early stopping. The classifier loaders apply amplitude scaling and synthetic noise, without random cropping or flipping.

The noise evaluation adds fresh AWGN at SNRs of $40, 30, 20, 10, 5, 0, -5$, and -10 dB. These values describe the power ratio of the stored trace to the added noise. They are not estimates of the total physical SNR of an already noisy recording, and 40 dB is not a noise-free reference. Each model is evaluated with newly drawn noise, so equal SNR settings do not imply identical noise realizations across models.

The same validation subset used for checkpoint selection is reused for noise evaluation. In addition, the auxiliary pretraining script iterates over the full dataset before the classification split, without using class labels. Hence, the evaluation includes exposure to validation waveforms during auxiliary pretraining and reuse of validation data for model selection and reporting. The results below are reported validation measurements under this protocol; they are not estimates from an untouched test set. The originally reported computing environment was an AMD Ryzen 9 7950X CPU, an RTX 4070 Ti Super GPU, Python 3.9, PyTorch 2.1.0, and CUDA 12.2. This environment is taken from the original experimental record and has not been independently verified.

C. Metric Definitions

Accuracy is the fraction of correctly classified single-channel examples (channel-windows). The evaluation functions compute precision, recall, and F1 using an unweighted macro average over classes. For class c,

$$p_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \quad r_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \quad f_c = \frac{2p_c r_c}{p_c + r_c}, \tag{13}$$

In (13), undefined values are set to zero. Each macro metric is the arithmetic mean of its six class-specific values. In particular, macro F1 is not computed as the harmonic mean of macro precision and macro recall. Macro averaging is relevant to the unequal class counts in Table I. It remains a six-class classification metric, not an event-versus-background detection rate or a false-alarm rate.

TABLE II
IMPLEMENTED COMPARISON ARCHITECTURES.

Plot label	Local architecture	Parameters
TimeMIL	1D CNN (32/64/128 channels), gated MIL pooling	57,415
InceptionTime	Two Inception modules; kernels 9/19/39 and pooling branch	153,991
MSBA	Parallel kernels 5/11/21, bidirectional GRU, temporal attention	53,510
CBiGRU-AT	1D CNN, bidirectional GRU, temporal attention	69,766
CLSTM-Attn	1D CNN, LSTM, temporal attention	114,822
PG-ZGCA	Two ResNet18 encoders, auxiliary CNN, gated cross-attention	23,411,114

D. Illustrative Auxiliary Representation

Fig. 2 compares the power spectrograms of a manual-knocking waveform from the example export and its auxiliary CNN output. It uses the transform in (7) and common color limits. The two representations differ in amplitude and spectral distribution, illustrating the learned signal transformation. Assessing force recovery or denoising accuracy would additionally require a calibrated observation model and an appropriate reference signal. The frequency-bin and frame-index axes retain the discretization used by the implementation without inferring instrument metadata from the integration interval.

E. Comparison Models and Noise Robustness

The comparison figures report results for five local baselines labeled TimeMIL, InceptionTime, MSBA, CBiGRU-AT, and CLSTM-Attn. These names refer to the local implementations summarized in Table II. In particular, the local TimeMIL implementation is a three-layer CNN with gated multiple-instance pooling, and the local InceptionTime implementation has two Inception modules. These simplified implementations have not been verified as reproductions of the published methods. The labels in the archived plots identify the local models in Table II; conclusions are restricted to those implementations.

The comparison models use waveforms directly, whereas PG-ZGCA uses two spectrogram encoders and a pretrained auxiliary CNN. They also differ substantially in parameter count and pretraining exposure. Their shared classifier training settings facilitate comparison under the implemented protocol, but do not control these factors. Because the comparison includes convolutional and recurrent models, the observed noise sensitivity cannot be assigned to a common Transformer self-attention mechanism.

Figs. 3–5 present the archived single-run validation curves. Scores declined as the added SNR decreased, and PG-ZGCA had higher reported scores than the five plotted local models. At -10 dB, the full model's tabulated accuracy is 0.9297, with macro precision 0.7988, macro recall 0.7319, and macro F1 0.7409. These point estimates and curves are taken from the original results; a new full-dataset evaluation was not performed. Prediction-level logs are unavailable for reconstructing uncertainty estimates or confusion matrices. The curves alone do not demonstrate that uncertain examples are preferentially assigned to no event, nor do they measure an operational missing-alarm rate.

TABLE III
REPORTED VALIDATION RESULTS OF ARCHITECTURAL VARIANTS AT -10 DB ADDED SNR.

Model	Acc.	P	R	F1
PG-ZGCA	0.9297	0.7988	0.7319	0.7409
A: Raw branch	0.9236	0.7717	0.6529	0.6894
B: Direct addition	0.9256	0.7817	0.6404	0.6783
C: Fixed attention	0.9243	0.7486	0.6547	0.6799

F. Architectural Variants

Three variants accompany the full model. Variant A removes the auxiliary representation, the second ResNet18 encoder, attention, and layer normalization; it pools the raw encoder features directly. Variant B keeps both encoders and replaces attention with direct feature addition, $X = \mathrm{LayerNorm}(R + P)$. Variant C retains attention and normalization but replaces the learned gate with a fixed attention contribution, $X = \mathrm{LayerNorm}(R+A)$. All four classifier definitions use dropout 0.5 and a six-class linear output layer.

Table III summarizes the reported validation results. The full model exceeded A by 0.61 percentage points in accuracy, 7.90 points in macro recall, and 5.15 points in macro F1. Its macro F1 exceeded B and C by 6.26 and 6.10 points, respectively. These are descriptive differences from one reported run, rather than significance-tested effects.

Checkpoint differences further limit interpretation of these comparisons. The auxiliary CNN weights embedded in the full checkpoint differ from those embedded in variants B and C and from the separately archived auxiliary checkpoint. Although the training definitions freeze the auxiliary CNN, the archived variants do not establish a shared auxiliary initialization. Variant A also changes capacity and normalization, while B removes both attention and its gate. The table compares archived configurations, with module-specific attribution limited by these differences.

G. Interpretation and Limitations

PG-ZGCA provides an explicit route for an auxiliary signal prior to influence feature selection. The exact initialization property is the zero attention contribution in (11); the remaining normalized raw path is still trained for classification. The learned gate adjusts the attention contribution. Its magnitude and the frozen status of the auxiliary CNN provide architectural information, rather than independent measures of physical reliability, explanation quality, or ground-truth force.

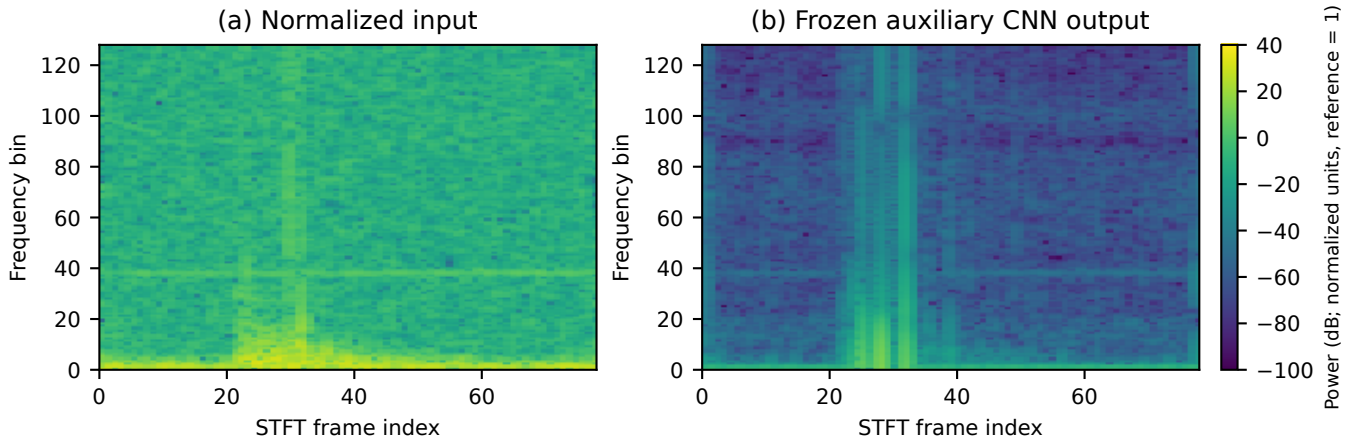


Fig. 2. Power spectrograms of an illustrative manual-knocking waveform and its auxiliary CNN output. Both power spectrograms use the same transform, reference, and color limits. The auxiliary output is computed using the CNN embedded in the archived full-model checkpoint. The example is a stored normalized recording without newly injected AWGN; the panels illustrate a representation change rather than a quantitative comparison with clean ground truth.

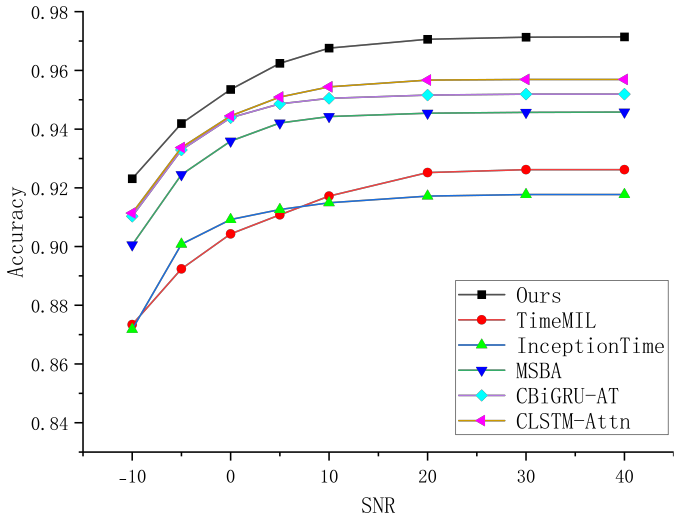


Fig. 3. Reported single-run validation accuracy under added AWGN. Baseline names identify the local implementations in Table II. The archived curve is retained; independent test results and uncertainty estimates are not available.

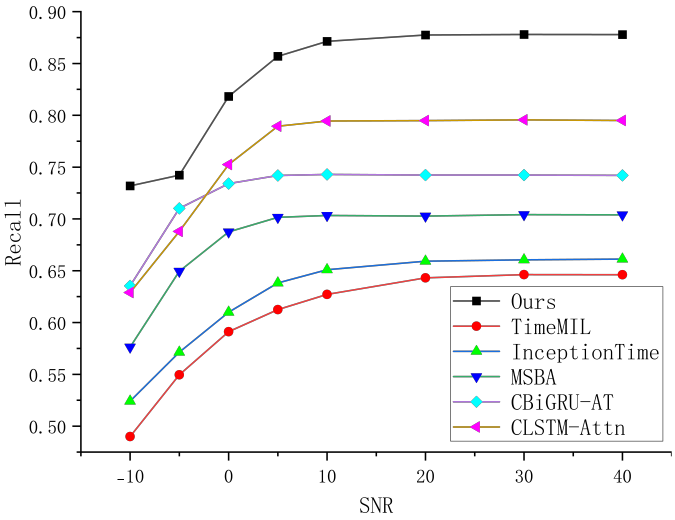


Fig. 4. Reported macro recall under added AWGN on the reused validation subset. This is the average six-class recall, including no event, rather than a separately measured event detection probability.

The present evaluation has four main limitations. First, sample-level random splitting can share recording conditions and related channels across subsets; recording-grouped evaluation is needed to assess generalization. Second, auxiliary pretraining on the entire collection and validation reuse limit independence of the reported scores. Third, heuristic labels, unequal parameter counts, differing auxiliary checkpoints, and a single run limit causal attribution. Fourth, synthetic AWGN does not represent every form of environmental interference or sensor drift. Calibration of the measurement-to-dynamics mapping, training-only auxiliary pretraining, matched-checkpoint variants, repeated runs, and an untouched recording-grouped test set are needed before making claims about deployment reliability, false alarms, or generalization to new sensing installations.

V. Conclusion

This paper presents PG-ZGCA, a DAS event classifier that combines a dynamics-inspired auxiliary CNN with independent spectrogram encoders and zero-initialized gated cross-attention. The auxiliary target is formed by detrending and discrete integration of normalized recordings, and serves as an equivalent signal prior. Auxiliary features provide attention queries, while raw features provide keys, values, and the residual path. At initialization, the attention contribution is zero and the fused features equal the layer-normalized raw features.

On the available six-class collection, the reported single-run validation result at -10 dB added SNR is 0.9297 accuracy

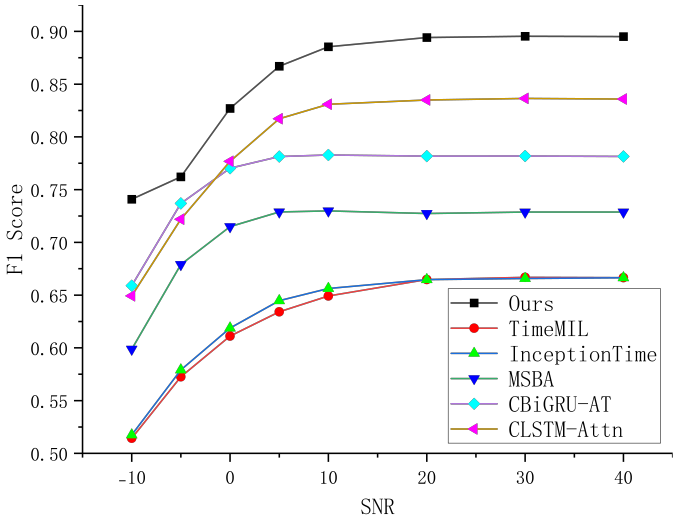


Fig. 5. Reported macro F1 under added AWGN on the reused validation subset. The baseline architecture definitions and evaluation limitations are shared with Figs. 3 and 4.

and 0.7409 macro F1. The results suggest that the combined representation is useful within the implemented evaluation protocol. Validation reuse, full-collection auxiliary pretraining, sample-level splitting, and checkpoint differences constrain stronger conclusions. Future evaluation should use calibrated sensor information, training-only pretraining, matched model variants, and independent recording-grouped test data to establish the contribution and transferability of the physical prior.

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